

Detecting sniffers

As mentioned earlier, since sniffers work silently, it is very difficult to detect them on a network. There are, however, a few tricks that can provide a clue to a possible sniffer presence. There are two ways to detect a sniffer, one being host-based and the other network-based. In host-based detection, you can use small utilities to detect if the NIC is running in a promiscuous mode on any host in a network. Since the basic requirement for a sniffer to work is to put the network interface in 'read all' mode, disabling it can very effectively help shutting down stray sniffers. In case of network-based detection, anti-sniffer software can be run to detect the presence of specific signature packets. In another approach, scripts can be run to check each network host for the presence of known sniffers, processes, etc. Modern anti-virus or anti-spyware software are capable of detecting sniffing software and disabling it.

Protection from sniffers

While the very first step should be to design a tight perimeter defence system while creating network architecture, there are a few methods that could be deployed to make the infrastructure less sniffer-prone. The following tricks help achieve that to a great extent.

Disabling promiscuous mode on network interfaces results in shutting down most sniffer software. This can be done by running an admin script as a daily job on the network, or deploying a network policy at the host level to control access to the network card configuration settings.

Using switched networks can reduce the possibility of a sniffer running on the network. Unlike in a network hub, in a switched network the packets are delivered to the destination and are not visible to all nodes—thus reducing the chances of someone sniffing it on the way. Also, for network administrators it becomes easy to detect sniffers by focusing on the switched network segments, one at a time.

Anti-sniffing tools can be used to detect the network interface mode, as well as various processes and software present on servers or network hosts. Modern intrusion-detection systems have this as an integrated feature.

IPSec encryption can be used for token-based packet security in the network infrastructure, if the data is of a confidential nature. IPSec provides data encapsulation and encryption of high standards, and is available on modern routers, firewalls and other network components. Almost all operating systems do support IPSec, and it is widely used in serious IT infrastructure. For session layer protection, SSL and TLS can be used to encrypt traffic.

Protecting FOSS systems

Let us now look at a few sniffer products, in order to learn about what is used in the FOSS world today. Linux systems use the *tcpdump* utility, which is an excellent built-in sniffer to capture and store TCP packets. As for third-party open source tools, Wireshark (Ethereal) is very famous due to its GUI interface, and packet-filtering and viewing capabilities. Sniffit, DSniff and Ettercap are similar utilities, but meant for different purposes. DSniff is powerful in terms of capturing SSL traffic.

FOSS systems have no built-in method to protect themselves from sniffers. The methods described above could pretty much be used for various Linux distros, to make those less vulnerable to sniffer attacks. A smart utility available on Linux distros, called AntiSniff, can be used in a script to detect network interfaces in promiscuous mode.

Network sniffing is difficult to detect because it is a passive and silent type of attack. There are methods to detect and disable sniffers, and network administrators should incorporate those into their network to protect their IT infrastructure from information loss or theft. 

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